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June 10, 1965 and
INTERIM REVISION OF
Fed. Spec. GG-P-146A
June 25, 1956

INTERIM FEDERAL SPECIFICATION

PEN SET, DESK

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D.C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification GG-P-146A, dated June 25, 1956.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers desk pen sets of the dip, fountain, and ball point pen types; and replacement pen point and feed section units for the dip and fountain pen sets. Each set consists of a pen and a base.

1.2 Classification.

1.2.1 Types. The pen sets shall be of the following types, as specified (see 6.2):

Types:
- I - Dip type.
- II - Fountain pen type.
- III - Ball point pen type.

1.2.2 Pen point and feed section units for types I and II pens, and point of refills for type III ball point pens shall be furnished as specified (see 3.6, 3.7, and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

GG-B-60 - Ball Point Pen, Single Cartridge
TT-I-563 - Ink, Writing, for Fountain and Dip Pens.
UU-P-21 - Pad, Writing Paper.
UU-P-553 - Paper, Wrapping, Tissue.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil agencies).
Fed. Test Method Std. No. 191/GEN (General Provisions) - Textile Test Methods.
Fed. Test Method 191/2101 - Wool Content, Alkali Method.
Fed. Test Method 191/5041 - Determination of Weight of Textile Materials: Small Specimen Method.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Bid samples. Bid samples, when required by the invitation to bid, shall be submitted for each item offered in the quantities specified in the invitation (see 6.2), and shall be examined as specified in 6.3.

3.2 Materials. Materials shall be as specified herein and shall be of a quality which will insure compliance with the performance requirements of this specification. The materials shall be compatible with each other under all conditions encountered in the use of the pen sets.

3.2.1 Plastic. Plastic shall be of a quality, hardness, and rigidity suitable for the intended use, and shall have low moisture absorption under wide humidity and temperature conditions. The plastic shall be nonflammable, flame resistant, or slow burning when tested as described in 4.5.2. The plastic shall not have an objectionable odor, as received, nor in type I and II pen sets when in continuous contact with writing ink conforming to TT-I-563 (see 4.5.3). The plastic shall be ink-proof and shall not be subject to attack or deterioration by the ink nor the ink by the plastic when subjected to the test specified in 4.5.3. The plastic shall retain its design contours without warping, crazing, cracking or discoloring while in storage or service. The plastic shall, when subjected to the accelerated aging test described in 4.5.1, remain straight and rigid, without warping and shall show no change in appearance or dimensions which will affect serviceability.

3.2.2 Metals. Metals, except those concealed within the base which do not come into contact with ink, shall be corrosion resistant and shall show no evidence of corrosion when tested as described in 4.5.4. The point for pens of types I and II pen sets shall be of stainless steel, with a welded alloy tip (nib) to resist wear and meet the writing qualities specified herein. Pen points and welded alloy tips shall show no visible damage, such as cutting into or wearing away of the nib when tested as described in 4.5.5. The point may be treated or plated in accordance with standard commercial practice.

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3.2.3 Felt. The felt shall be a thoroughly-cleaned mixture of not less than 40 percent wool, reprocessed wool, or reused wool, or any combination thereof, and the balance vegetable or synthetic fibers (see 4.5.6). The weight per square yard of the felt shall be not less than 4.0 ounces (see 4.5.7).

3.3 Color. Base shall be black in color. Pen barrels shall be black or a combination of clear plastic and black.

3.4 Dimensions.

3.4.1 Base. If round, the area of the base shall be not less than 8 square inches and not more than 16 square inches. If other than round, no dimension shall be less than 2-1/2 inches and the area of the base shall be not less than 12 square inches and not more than 20 square inches. Exclusive of hinged socket, the height of the base shall not exceed 3 inches.

3.4.2 Pen assembly. The pen assembly including the pen point and feed section unit (types I and II) or the writing tip (type III) shall not exceed 7-1/2 inches in length and shall be not less than 5-3/4 inches in length. The pen assembly shall have an outside diameter of not less than 0.360 inch and not more than 0.495 inch at the maximum diameter of the barrel.

3.5 Assembly and replacement. Each pen set shall be designed to permit satisfactory assembly and replacement of replaceable component parts, such as pen point and feed section units, ink cartridge refills, without the use of tools.

3.6 Pen point and feed section unit. Types I and II pen point and feed section units shall be furnished, as specified, in the manufacturer's "fine," "medium," and "broad" grading, which represents the limits of written line generally accepted in the commercial field. Other designations such as "extra fine" and "stub" shall be furnished when specified. Point slits shall be centered with equal width of prong on each side. Slits shall be set with sufficient opening to permit a balanced ink flow but not so wide as to stop capillary action. Points shall be parallel, not one up and one down. Each pen point shall be so finished and polished or ground that it will write smoothly in all directions when the pen is held at an angle of 50° to 75° from the horizontal, and that it will not pick up particles of paper nor dig into paper while writing (see 4.5.8). Unless otherwise specified, types I and II pen sets shall be furnished with "medium" point and feed section units (see 1.2.2 and 6.2). Unless otherwise specified, type III pen sets shall be furnished with refills having "medium" points and black ink (see 1.2.2 and 6.2).

3.7 Replacement pen point and feed section units. Replacement pen point and feed section units for types I and II pen sets, conforming to this specification, shall be furnished individually in quantities and gradings as specified (see 1.2.2 and 6.2).

3.8 Interchangeability. All parts having the same manufacturer's part number shall be interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be made if the design is changed and interchangeability is affected.

3.9 Pen sets.

3.9.1 Base. The base shall be of glass, metal, ceramics, plastic, rubber, or a combination of these materials. The base shall be of stable construction and shall be provided with a pad or feet which will not mar a desk top. The pad or feet shall have non-skid properties and be made of natural or synthetic rubber, cork, a combination of rubber and cork, or felt. Vacuum-type base pads or pads with a pressure-sensitive adhesive will not be acceptable. The non-skid type pad or feet shall provide the base with an appreciable resistance to lateral pressure and shall provide no resistance to vertical lift. The base without pen assembly shall weigh not less than 6 ounces. The base shall be designed in such a manner that its parts will not become separated and it will not break or become deformed when tested as described in 4.5.9.

3.9.2 Pen assembly. The pen assembly sh . be of such design that it will fit the base socket snugly.

3.9.3 Barrel. The barrel shall be of plastic, metal, or a combination of these materials, and shall be smoothly finished. The barrel shall be round and the center line of the barrel shall be straight. The barrel shall be smoothly tapered from the maximum diameter to form a comfortable grip at the writing end, and from the maximum diameter to the non-writing end to form a smooth rounded tip. The barrel shall consist of two sections. The sections shall be provided with matching external and internal threads; which when joined, shall form a rigid barrel. Threads shall be of such size and pitch to produce smooth action in joining

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barrel sections and withstand ordinary conditions of use. Where strength of the joint is dependent upon the threaded parts only, the threaded portion of each section shall be a minimum of 1/4-inch. Where strength of the joint is dependent upon unthreaded parts in addition to threaded parts for the purpose of maintaining rigidity and other performance functions, the manufacturer's standard length of threaded parts will be acceptable provided the barrel meets the requirements specified herein. When both sections of the barrel are made of plastic, the barrel shall be provided with a metal band not less than 1/16-inch in width. The finished thickness of the band material shall be not less than 0.007 inch. The band shall be flush with both sect[illegible] of the barrel.

3.10 Shelf life. Pen sets, under normal storage conditions, shall retain their serviceability as defined by this specification for not less than 12 months from date of shipment. Suppliers shall make replacement, without cost to the Government, of all pen sets failing to meet these requirements.

3.11 Identification marking. Each pen set shall be marked with the manufacturer's name or trade-mark. The barrel shall be legibly and permanently stamped or marked with "U.S. Government" in letters of durable, contrasting color of 1/16-inch minimum and 1/8-inch maximum height. Each type I and type II pen point shall be marked to indicate grading, such as "fine," "medium," etc., by the manufacturer's part number or grading to identify the pen point furnished therewith.

3.12 Finish. The pen sets shall be finished in accordance with standard commercial practice.

3.13 Workmanship. Workmanship shall be of the highest grade to produce pen sets and component parts [illegible]table for the intended purpose and in accordance with standard commercial practice. The pen sets, including all component parts, shall be free from burrs and sharp or rough edges. Barrels shall be free from defects such as mold fins, cold bonds, delaminations, porosity, warpage, checks, cracks, chipped edges, blisters and scratches. Threads shall be true to form, cleanly cut, and free from defects which might impair serviceability. Particular attention shall be given to concentricity, formation, fit and mating of parts, and accuracy of dimensions. The pen sets shall be free from any defects not specified which effect appearance or which might affect serviceability.

3.14 Type I, dip type.

3.14.1 Component parts. Type I pen sets shall consist of a dip pen assembly and a base with an ink reservoir in a dip well to support the pen assembly and to ink the pen point and the ink-retaining feed behind the point.

3.14.2 Base. Type I base shall have a built-in refillable-type ink container. The base shall include a socket which shall be designed to hold the pen snugly at an angle between 45° and 70° from the horizontal. The base shall be designed in such a manner that ink will not spill when the base is turned upside down (see 4.5.10).

3.14.3 Reservoir. Type I shall have an expansion chamber and reservoir constructed so that (a) the pen point and feed section of the pen remain in the ink to a level of approximately 3/8 inch from the end of the writing point when the reservoir is one-quarter or more full or (b) the point and feed section of the pen rest against the small opening in the ink channel through which the ink flows from the main ink reservoir, or (c) it contains a long lasting capillary device, constructed to insure a uniform, adequate supply of ink to the point and feed section at all levels of the ink reservoir. The capacity of the reservoir shall be a minimum of one ounce. The reservoir shall be such that it can be easily cleaned without the use of special tools, and refilled.

3.14.4 Pen assembly. Type I assembly shall consist of a barrel, a pen point, and a feed section.

3.14.5 Barrel. The writing end of the barrel shall be provided with internal threads to match the external threads of the pen point and feed section unit, and a shoulder to provide a seal when the pen point and feed section unit is installed.

3.14.6 Pen point and feed section unit. The pen point and feed section unit shall consist of a pen point, an ink-retaining feed, and a sleeve which holds the pen point and feed together. The pen point and ink-retaining feed shall be permanently pre-assembled in the sleeve as a compact unit in a manner to preclude the need for any additional adjustment. The sleeve shall be provided with external threads to match the internal threads of the barrel. The assembled unit shall conform to the configuration and dimensions specified in figure 1. All dimensions shown in figure 1 are in inches.

3.14.7 Design and performance. Type I pen sets shall be designed to permit a minimum of evaporation of ink and allow a controlled ink level at the base. Each set shall efficiently use at least 75 percent of the ink that it contains when the reservoir is filled to the proper level. The pen point and section feed unit shall fit the barrel snugly by means of a threaded assembly. The pen set shall be so designed that the finger grip of penholder remains clean in normal use and in a 40°F. temperature rise. Ink feeding shall be of the self-feeding type which permits continuous writing with infrequent dipping.

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Pen points shall not drip ink. Pen sets shall operate efficiently under any temperature within the range of plus 30° to plus 120°F. The pen sets shall operate efficiently without spillage or overflow with a rise or fall in temperature of 40°F. in 2 hours when the reservoir is filled to normal capacity. The pen sets shall operate efficiently on a continuous or intermittent basis. Pen point and feed section unit shall fit snugly in the proper position in holders. (See 4.5.11.)

3.14.8 Ink utilization. Unused pen point and feed section units, or those which have been unused for a sufficient period of time to become dry, shall accept 60 to 80 milligrams of ink within 1 minute after inserting the pen in the socket of a reservor which has been filled to capacity, when tested as described in 4.5.12. Pen point and feed section units as described and under the same conditions, shall accept 90 percent of the specified quantity of ink, when the ink reservor is one-fourth filled (see 4.5.12.)

3.15 Type II, fountain pen type.

3.15.1 Component parts. Type II pen sets shall consist of a fountain pen assembly and a base with a socket designed to hold the pen with a minimum evaporation of ink.

3.15.2 Base. Type II base shall be provided with a socket for holding the pen. The socket shall be attached to the base and hold the pen snugly at an angle between 45° and 70° from the horizontal or may be hinged to allow a variable angle.

3.15.3 Pen assembly. Type II pen assembly shall consist of a barrel, and a pen point and feed section unit.

3.15.4 Barrel. The barrel shall have an efficient self-filling mechanism of the lever or plunger type, or other filling device which will provide equivalent quality and performance when used with pen point and feed section unit specified in 3.14.6. Lever-type mechanisms shall have a double-action pressure bar. The filling device shall draw into the ink tube not less than between 0.75 milliliter of ink when operated according to manufacturer's instructions for filling. The filling device shall be so designed that not more than four strokes or pressures are required for filling of the quantity specified. The writing end of the barrel shall be provided with internal threads to match the external threads of the pend point and feed section unit, and a shoulder to provide a seal when the pen point and feed section unit is installed.

3.15.5 Ink tube. The ink tube or sac shall have a capacity of not less than 1 milliliters of ink. The ink tube shall be of natural rubber, plastic, or material of equivalent strength and durability in accordance with that acceptable in standard commercial practice. The material shall not be subject to attack or deterioration by ink nor the ink by the material.

3.15.6 Pen point and feed section unit. The pen point and feed section unit shall be as specified in 3.14.6 for type I pen sets.

3.15.7 Design and performance. Each type II pen set shall be designed so that the finger grip of the pen remains clean during normal use and in a 40°F. temperature rise and fall. Pens shall operate efficiently under any temperature within the range of plus 30°F. to plus 120°F. The pen sets shall operate efficiently on a continuous or intermittent basis and pens shall not drip ink when in normal use. (See 4.5.13.)

3.16 Type III, ball point pen type.

3.16.1 Component parts. Type III pen sets shall consist of a ball point pen assembly and a base.

3.16.2 Base. Type III base shall be as specified in 3.15.2 for type II pen sets.

3.16.3 Pen assembly. Type III pen assembly shall consist of a non-retractable type barrel (may include spring or springs) and a replaceable refill (ink cartridge).

3.16.4 Barrel. The barrel shall be designed and constructed to fit and function satisfactorily with the refill, specified in 3.16.5. The barrel shall be so constructed as to provide continuous equalization of air pressure between the non-writing end of installed refill and the outside of the barrel. At least 0.100 inch of the refill writing tip, specified herein, shall project from the barrel. There shall be no permanent set, breaking, or cracking of barrel when tested as described in 4.5 . When made of plastic, the writing end of the barrel shall be provided with a metal tip. The outer surface of the metal tip shall be flush with the plastic, shall be tapered, and shall extend beyond the plastic at least 1/4-inch. The metal tip shall be securely and permanently attached to the plastic in such a manner that the barrel will meet the performance requirements specified herein.

3.16.5 Refill. The refill shall conform to GG-B-60.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that suppliers and services conform to prescribed requirements.

4.2 Lot. For purposes of sampling, a lot shall consist of all pen sets of the same type and pen point grading (types I and II); pen sets with refills of the same point size and color of ink (type III); and pen point and feed section units of the same grading offered for inspection at one time.

4.3 Sampling, inspection levels, and acceptable quality level.

4.3.1 Sample units.

4.3.1.1 Visual examination. The sample unit shall be one pen set or one pen point and feed section unit.

4.3.1.2 Dimensional examination. The sample unit shall be one pen set or one pen point and feed section unit.

4.3.1.3 Tests. The sample unit shall consist of the pen sets required to conduct all of the specified tests one time, or the pen point and feed section units required to conduct all of the specified tests one time.

4.3.2 Sampling for acceptance. A random number of sample units shall be taken from each inspection lot to form the required sample size for examinations and tests in accordance with MIL-STD-105.

4.3.3 Sampling for preparation for delivery. Sampling shall be in accordance with MIL-STD-105. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, and selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 4.0 defects per hundred units.

4.3.4 Inspection levels and acceptable quality levels (AQL's). The inspection levels and AQL's expressed in percent defective shall be as follows:

Characteristic	Inspection level	AQL	
		Major	Minor
Pen sets:			
Visual examination	I	2.5	4.0
Dimensional examination	S-3	---	4.0
Tests (all tests combined)	S-1	6.5	---
Pen point and feed section units:			
Visual and dimensional examination	I	2.5	---
Tests (all tests combined)	S-1	6.5	---

4.4 Examinations. Samples taken in accordance with 4.3.2 and 4.3.4 shall be examined for defects listed in tables I and II. The Government reserves the right to classify as a defect any departure from the specification not listed in tables I and II.

TABLE I. Classification of defects of pen sets

Defects	Category	
	Major	Minor
Visual:		
Not type specified.	X	
Pen point and feed section unit not of the grading specified (types I and II).		X
Point not of the size specified (type III).		X
Materials not as specified.	X	

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TABLE I. Classification of defects of pen sets (con)

Defects	Category	
	Major	Minor
Color not as specified in 3.3		X
Tools required for replacement of component parts.		X
Point slits not centered (width of prongs not equal).		X
Points not parallel.		X
Base not as specified in 3.9.1.		X
Pen assembly does not fit base socket snugly.	X	
Barrel not as specified in 3.9.3.	X	
Identification marking not in conformance with 3.11.		X
Workmanship not as specified in 3.13.	X	
Type I:		
Components not as specified in 3.14.1.	X	
Base not in accordance with 3.14.2.		X
Reservoir not in conformance with 3.14.3.		X
Pen assembly not as specified in 3.14.4.		X
Barrel not provided with internal threads and shoulder as specified in 3.14.5.	X	
Pen point, ink-retaining feed or sleeve missing.	X	
Pen point, ink-retaining feed and sleeve not permanently pre assembled.	X	
Sleeve not provided with external threads specified in figure 1.	X	
Configuration of pen point and feed section unit not in conformance with figure 1.	X	
Type II:		
Components not as specified in 3.15.1.	X	
Base not in conformance with 3.15.2.		X
Pen assembly not as specified in 3.15.3.		X
Barrel not in conformance with 3.15.4.	X	
Ink tube not in conformance with 3.15.5.		X
Pen point, ink-retaining feed or sleeve missing.	X	
Pen point, ink-retaining feed and sleeve not permanently pre-assembled.	X	
Sleeve not provided with external threads specified in figure 1.	X	
Configuration of pen point and feed section unit not in conformance with figure 1.	X	
Type III:		
Components not as specified in 3.16.1.	X	
Base not in conformance with 3.16.2.		X
Pen assembly not as specified in 3.16.3.		X
Barrel not in conformance with 3.16.4.	X	
Refill not furnished with point size specified.		X
Refill not furnished with color of ink specified.		X
Dimensional:		
Dimensions of base not as specified in 3.4.1.		X
Dimensions of pen assembly not as specified in 3.4.2.		X
Dimensions of pen point and feed section unit not in conformance with figure 1 (types I and II).		X

TABLE II. Classification of defects of pen point and feed section units

Defects	Category	
	Major	Minor
Visual and dimensional:		
Materials not as specified.	X	
Pen point and feed section unit not of the grading specified.		X
Point slits not centered (width of prongs not equal).		X
Points not parallel.		X
Pen point, ink-retaining feed or sleeve missing.	X	
Pen point, ink-retaining feed and sleeve not permanently pre-assembled.	X	
Sleeve not provided with external threads specified in figure 1.	X	
Configuration of pen point and feed section unit not in conformance with figure 1.	X	
Dimensions of pen point and feed section unit not in conformance with figure 1.	X	

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4.5 Tests. Samples taken in accordance with 4.3.2 and 4.3.4 shall be subjected to the tests specified in table III. Any sample unit failing to pass one or more tests shall be considered defective. Unless otherwise specified herein, tests shall be conducted in an atmosphere maintained at 23° ± 2° C. and a relative humidity of 50 plus or minus 5 percent. When type III pen assemblies are furnished with a spring or springs (3.16.3), all tests shall be conducted with the spring or springs in the pen barrels. A pen barrel and case, conforming to the requirements of this specification, shall be used in testing replacement pen point and feed section when ordered separately.

TABLE III.

Tests	Applicable paragraph	Types I	Types II	Types III	Pen point and feed section units
Flammability	4.5.2	X	X	X	
Deterioration	4.5.3	X	X		X
Corrosion resistance	4.5.4	X	X	X	X
Abrasive	4.5.5	X	X		X
Wool content of felt	4.5.6	X	X	X	
Weight of felt	4.5.7	X	X	X	
Writing	4.5.8	X	X		X
Drop	4.5.9	X	X	X	
Spill	4.5.10	X			
Temperature (type I)	4.5.11	X			
Ink utilization	4.5.12	X			
Temperature (type II)	4.5.13		X		
Breaking test	4.5.14	X	X	X	

4.5.1 Accelerated aging test for plastic. The samples shall be subjected successively to each of the following conditions and examined for compliance with 3.2.1:

24 hours at 160° ± 2°F. in an oven with the specimen placed horizontally on an insulated shelf at 70 - 75% relative humidity.

24 hours in a cold chamber at -40° ± 4°F., starting within 30 minutes after the first step of the exposure cycle has been completed.

24 hours at 160° ± 2°F. in an oven starting within 2 hours after the second step of the exposure cycle has been completed.

4.5.2 Flammability test for plastic. A draft shield shall be used to reduce air current effects. Hold the specimen horizontally within the shield above a Bunsen burner having a 1-inch to 1-1/4-inch blue flame just touching the end of the plastic until the specimen ignites, or for a period of not over 30 seconds. Remove from the flame. If the specimen will not support combustion, it is nonflammable. If the specimen burns when in contact with the flame but will not continue to burn when removed from the flame, it is flame resistant. If the specimen continues to burn at a rate of less than 2.5 inches per minute, it is slow-burning. If the specimen bursts into flames immediately, it shall be considered as not meeting the requirements of this specification. (See 3.2.1.)

4.5.3 Deterioration test. Fill the ink reservoir of type I pen set or the ink tube of type II pen set with writing ink conforming to TT-I-563. Allow pen to stand for 15 days at a temperature of 140°F. with pen continuously in socket. Examine pen and ink for compliance with 3.2.1. A low-power microscope with magnification over a range of 2x to 4x shall be used for examination of deterioration.

4.5.4 Corrosion resistance test. Metal parts shall be placed in a boiling 10 percent (by weight) aqueous solution of sodium chloride for a period of fifteen minutes. Metal parts upon being removed from this solution shall be immediately immersed for at least one hour in 10 percent (by weight) aqueous solution of sodium chloride at room temperature. They shall then be removed from this solution and, without having the adhering liquid wiped off, allowed to dry for 24 hours at room temperature and then examine for presence of corrosion. (See 3.2.2.)

4.5.5 Abrasive test. Lay a 6-inch double cut smooth or second cut file on a flat surface. Exert 2 ounces of pressure and draw the pen point and feed section unit the full length of the file in an approximate writing angle. Examine for compliance with 3.2.2.

4.5.6 Wool content of felt test. The wool content of the felt shall be determined as described in method 2101 of Fed. Test. Method Std. No. 191.

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4.5.7 Weight of felt test. The weight per square yard of the felt shall be determined as described in method 5041 of Fed. Test Method Std. No. 191.

4.5.8 Writing test. Ink conforming to type I of TT-I-563 and paper conforming to UU-P-21 shall be used for this test. Dip sample pen assembly into ink as frequently as necessary. Move pen at an angle of 50° to 75° from the horizontal across a sheet of paper. On the same paper write "Pen Set, Desk" 15 times. Examine for compliance with 3.6.

4.5.9 Drop test. Perform this test by dropping a sample base from a height of 30 inches onto an asphalt-tile concrete floor. The tile shall be 1/8 ± 1/64 inch thick, and shall be cemented to the concrete in accordance with the application instructions of the tile manufacturer. Support the sample with a press clamp so that the bottom of the base strikes the floor. Examine for requirements of 3.9.1.

4.5.10 Spill test (type I). The base, filled with ink to capacity, shall be turned upside down for a minimum of 10 seconds and then returned to upright position. Repeat the foregoing twice. Examine for requirements of 3.14.2.

4.5.11 Temperature test (type I). Each type I pen set, assembled with reservoir filled to normal capacity with ink conforming to TT-I-563, shall be subjected to successive temperature changes and examined in accordance with table IV. (See 3.14.7.)

TABLE IV.

Period	Temperature	Examine for
Hour	°F.	
2	30	Applicable requirements of 3.14.7.
2	70	Overflow and spillage of ink.
2	30	Operating efficiency.
2	120	Applicable requirements of 3.14.7.

4.5.12 Ink utilization test (type I). Weigh and record the weight of an unused type I pen assembly in dry condition. Insert the pen assembly for a period of 1 minute in the socket of ink reservoir filled with writing ink to capacity. Withdraw the pen with picked up ink and weigh again to determine and record the amount of ink picked up by the pen point and feed section. Repeat the foregoing procedure with a pen which has been wetted and in use but without sufficient ink remaining in the feed and pen point to produce any writing, leaving the pen inserted in the socket of ink reservoir as above specified, for a period of 15 seconds. Withdraw the pen with picked up ink and weigh to determine and record the amount of ink picked up by the pen point and feed section. Repeat the entire procedure with the ink reservoir one-fourth filled. Examine recordings for compliance with 3.14.8.

4.5.13 Temperature test (type II). Fill sample pen assembly with 0.75 milliliter of ink conforming to TT-I-563. Wipe the ink from the pend assembly with lint free cloth or paper. Operate the filling device so that one drop of ink falls from the pen point to the bottom of a clean test tube (10 millimeters by 100 millimeters), taking care that the ink drop does not touch the interior wall of the test tube. Insert a cotton swab (Q-Tip or equal) into the drop of ink in the test tube. Retain this swab for comparison purposes. Wipe the ink from the pen assembly with lint free cloth or paper. Press the cloth or paper against the hole in the convex side of the pen point until ink is removed from the collector grooves by capillary action. Insert the pen assembly in the base socket of the sample. Expose the sample pen set for two hours at each of the temperatures specified in table V. After each exposure remove the sample from test chamber and the pen assembly from the socket, exercising care that the pen point does not touch the interior wall of the socket. Then, wipe the interior wall and bottom of the socket with an unused cotton swab. Compare the color of the second cotton swab with the color of the swab inserted in the drop of ink at the bottom of the test tube. Discoloration of the second cotton swab to the degree exhibited by the swab inserted in the drop of ink, or evidence of ink on the fingergrip of the pen assembly shall be cause for rejection of the sample. Before each temperature exposure wipe the interior wall and bottom of the base socket with lint free cloth or paper. (See 3.15.7.)

TABLE V.

Period	Temperature	Examine for
Hour	°F.	
2	30	Applicable requirements of 3.15.7.
2	70	Leakage after 40° rise in temperature, and performance.
2	30	Leakage after 40° fall in temperature, and performance.
2	120	Applicable requirements of 3.15.7.

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4.5.14 Breaking test. A sheet of paper shall be placed on a platform scale. A pen assembly shall be held between 1-1/4 and 1-1/2 inches from the writing end and at an angle of 50 degrees from the horizontal without the hand touching the paper. A force of 6 pounds shall be applied gradually and uniformly, in order to avoid variations and shock, within a period of approximately 5 seconds and immediately released. Examine the pen assembly for requirements of 3.16.4.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with table VI.

TABLE VI. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Pen sets. Each pen set shall be packaged in a box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. Prior to packaging each base shall be wrapped with paper conforming to UU-P-553. The type I and II pens shall be disassembled and the points protected with a chipboard or paperboard tube. The type III pens shall not be disassembled. Each complete assembled or disassembled pen shall be placed in an end opening kraft, glassine, or cellophane envelope. The boxes shall be secured to prevent accidental opening during shipment.

5.1.1.2 Pen point and feed section units. Each unit shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665 or PPP-B-676 or in a paperboard or chipboard tube with a cap closure. The containers shall be secured to prevent accidental opening during shipment.

5.1.1.2.1 Intermediate packaging. Twelve units of one description packaged as specified in 5.1.1.2 shall be intermediate packaged in a box conforming to PPP-B-566, PPP-B-665 or PPP-B-676. The boxes shall be secured to prevent accidental opening during shipment.

5.1.2 Level B. The pen sets, and pen point and feed section units shall be packaged and intermediate packaged as specified in 5.1.1.

5.1.3 Level C. The pen sets, and pen point and feed section units shall be packaged in accordance with the suppliers standard practice, providing that this will insure protection for the packaged item during shipment and provide for safe delivery to its destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Pen sets. Pen sets of one type shall be packed in boxes conforming to PPP-B-636, class weather resistant, grade V3c. Each box shall be closed, waterproof sealed with tape and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.2.1.2 Pen point and feed section units. Units of one description shall be packed in boxes conforming to PPP-B-636, class weather resistant, grade V3c. Each box shall be closed, waterproof sealed with tape and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.2.2 Level B.

5.2.2.1 Pen sets. Thirty six (36) pen sets of one type, shall be packed in a box conforming to PPP-B-636, type CF, class-domestic. Closure shall be in accordance with the appendix to the box specification.

5.2.2.2 Pen point and feed section units. Twenty four (24) intermediate packaged of units of one description shall be packed in a box conforming to PPP-B-636, type CF, class domestic. Closure shall be in accordance with the appendix to the box specification.

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5.2.3 Level C. Pen sets, and pen point and feed section units packaged as specified in 5.1 shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the Unifrom Freight Classification rules and the National Motor Freight Classification rules.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special markings required by the contract or order, all interior packages and shipping containers shall be marked in accordance with Fed. Std. 123.

5.3.2 Military agencies. In addition to any special markings required by the contract or order, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Special markings. In addition to the markings required by 5.3.1 or 5.3.2, the unit packages and shipping containers of type III pen sets shall be marked with a lot or batch control number.

6. NOTES

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type of pen set required (see 1.2.1).
(c) Grading of replacement pen point and feed section units required (see 1.2.2, 3.6, and 3.7).
(d) Bid samples, if required, and quantity required (see 3.1).
(e) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(f) Selection of marking required (see 5.3).

6.3 Bid samples. When bid samples are required by the invitation to bid, they shall be examined for:

Balance of pen assembly.
Facility of use.
General "feel" of pen assembly.
Suitability of color of ink (type III only).
Workmanship.

6.4 A second crimp or bulge (see figure 1, GG-B-60) for use in automatic assembly operations is permissible in the refill, providing it has no adverse effect on functions or performance of pen or refill.

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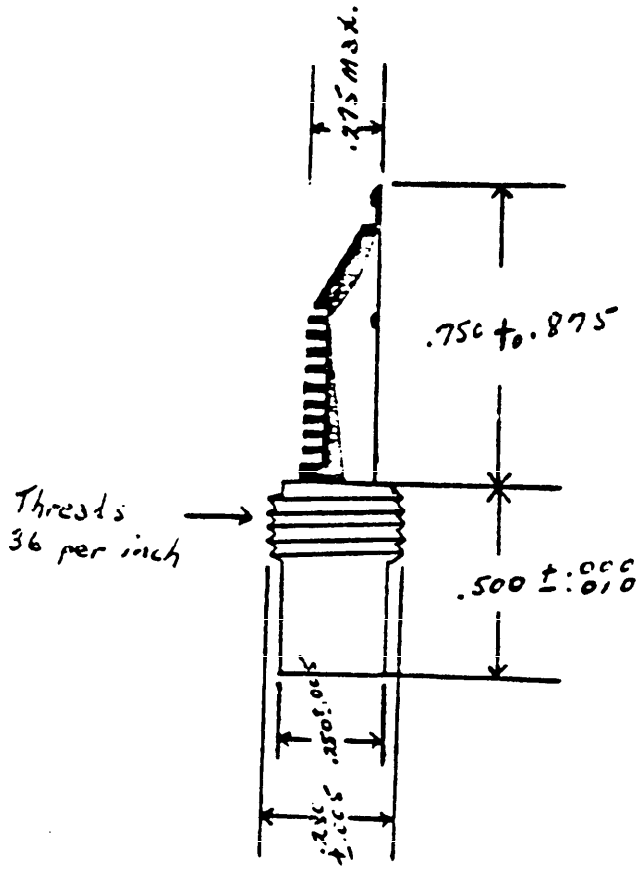


FIGURE 1.

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, and mail to: Gen. Services Admin., FSS (FMSO), Washington, D. C. 20406.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

GG-P-00146D (GSA-FSS) - PEN SET, DESK

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

FORM GSA FEB 69 2200

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